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Inventing New Road Paths for the Contact Theory Through Contact With a Puppet (a "Syrian Refugee Child") via a Teacher-Led Intervention in a Greek Primary School

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A pre–post quasi experimental design (test–retest–late response test) with a control group involving an intervention that aimed at creating contact conditions between the children of a classroom and a puppet "classmate" that held the identity of a Syrian refugee child was implemented in this study. Contact with a puppet is an innovative form of contact which combines features of direct and indirect intergroup contact. The sample comprised of 73 primary school students drawn from a relatively ethnically homogeneous primary school in Athens and attitudes measures, along with a behavioral measure (a money allocation task) were employed. Results showed that contact with the puppet improved children's attitudes toward the refugees, through inclusion of the other in the self, while an increase of empathy toward the refugees and a reduction of feelings related to threats posed by the refugees to the ingroup were also revealed. Results were more favorable in the intervention group that included contact and empathy inducing activities with the puppet. Most importantly, there were indications that some of the positive outcomes lasted in time, that is, 6 months after the intervention had finished, while there were also indications of the buffering effect for both intervention groups; again, more profound for the intervention that implemented empathy inducing activities.

Public Significance Statement

The study enhances intergroup contact through an intervention that implements contact with a puppet holding the identity of a Syrian refugee child with native children in a primary school in Greece. Contact with the puppet results in positive attitudes and contact intentions toward the Syrian refugee children. Empathy induction advances the effectiveness of the intervention in terms of resulting in long-lasting positivity (measured 6 months after) at a time of general public opinion deterioration of attitudes.

Keywords: intergroup contact, prejudice, threats, IOS, empathy

According to the United Nations High Commissioner for Refugees UNHCR, 2020 report, the world's forcibly displaced population had reached 79.5 million, that is, one in every 97 people, which constitutes a 100% increase since 2010 (41 million then vs. 79.5 million now). Fewer and fewer of those who flee are being able to return home (UNHCR, 2020). Accordingly, since 2015–2016 the European Union has been experiencing an unprecedented influx of refugees, with some particular European countries being at the

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frontline of receiving the majority of the refugee's wave. Greece is one of the host countries that have been accepting vast waves of refugees and asylum seekers, thus finding ways to integrate them is essential. A major challenge concerns the hard task of educating refugee children, who are mostly victimized by this situation. Another reason that highlights the importance of successfully integrating the refugee children in the Greek school is the potential of these children becoming the mediator between their families and the Greek school, eventually the Greek society.

Despite the urgent need for promoting positive intergroup relations in schools, and although prejudice-reduction interventions in schools based on indirect contact have been conducted for years, the majority of them has been conducted by researchers themselves and more rarely by school educators (Liebkind et al., 2018). They are also usually limited to 1-day, one-time interventions, thereby producing results which lack credibility in terms of persistence, after the intervention finishes (Hewstone et al., 2014; Paluck et al., 2019).

Theoretical Framework of the Study

The theoretical framework of the study is based on previous findings of Intergroup Contact studies and Prejudice reduction (e.g., Hewstone et al., 2014; Pettigrew & Tropp, 2006) and on findings coming from studies within the Integrated Threat Theory (Stephan & Stephan, 2000) as threats had been also used as a mediator between contact and prejudice, mainly in studies with adult populations. Findings from the field of *Symbolic Play* are also addressed, since there are implications and research evidence that indicate the potential benefits from using puppets for promoting empathy and perspective taking, through their perceived humanlike nature (Kroger & Nupponen, 2019). These findings are discussed in short in the following paragraphs.

The Contact Hypothesis for Promoting Intergroup Coexistence

A long line of research in the area of contact of diverse populations comes from the *Contact Hypothesis* originally proposed by Allport in 1954. In the decades following Allport's hypothesis research evidence has proven the impact of positive intergroup contact in achieving prejudice reduction through a procedure that includes reduction in perceived intergroup threats posed by the outgroup and anxiety, through increased empathy for the unprivileged outgroup or increased perspective taking and a sense of inclusion of other in the self (*IOS*), among other mediators. In fact, a variety of potential mediators has been studied and there are some strong indications of their impact, since they have repeatedly gained empirical support (for a general review please see Pettigrew & Tropp, 2006, 2008). These findings were observed in both benign and conflict or postconflict settings and regarded direct face-to-face contact (for a review please see Davies et al., 2011), indirect contact, for example, extended, imagined (for a review please see Zhou et al., 2019) and vicarious forms of intergroup contact (e.g., Liebkind et al., 2018; Vezzali et al., 2014). Overall, the most impressive outcome of Zhou et al.'s meta-analysis is that extended contact results in similar positive outcomes as those achieved by direct contact through friendship counterparts. These findings are of paramount importance, considering the ease of applying extended contact interventions, compared to direct contact interventions.

Different forms of indirect contact, like extended and imagined contact can either be seen on a continuum which aims at reducing prejudice and prepares the ground for the occurrence of actual, face-to-face positive intergroup contact (Crisp & Turner, 2012) or as distinct forms which may work through different mechanisms that subsequently have something unique to offer in fighting prejudice, thus being independently important (Turner et al., 2007). Regardless of the proven impact of either form of intergroup contact, a major shortcoming concerns the fact that the above findings have been mostly retrieved from cross-sectional studies and less from experimental or field studies, when the need for real-life evidence of causality, within naturalistic settings is imperative.

An Emotional Ace in Fighting Prejudice: Empathy (and Inclusion of Other in the Self)

A robust finding in the intergroup contact era regards the mediating role of empathy in contact situations for prejudice reduction (Pettigrew & Tropp, 2008; Vezzali et al., 2017; Zhou et al., 2019), while empirical evidence unravels the underlying mechanism of the way empathy results in prejudice reduction. In specific, empirical data coming from a meta-analysis of 58 articles show that empathetic concern and response are closely linked to perceiving a person as similar to oneself (Cotton, 1992), that is, *Inclusion of Other in the self*. This finding indicates the important role of creating contact situations, under optimal conditions, between unfamiliar groups of people, in order to first make the unfamiliar familiar (cognitive processing) and then to start acknowledging similarities and shared "properties," in other words starting perceiving the outgroup as part of oneself (or of the ingroup) and being able to stand in that person's (or group) shoes. In this sense, *Inclusion of Other in the Self* is conceptually related to empathy. Similarly, pioneering work by Bruneau et al. (2015) showed that narrations regarding the outgroup members' mental characteristics can dilute an emphasis on group identities and reduce the gap between ingroup and outgroup empathy.

Studies with children show that the ability to respond empathetically to another person's needs develops with age, since cognitive procedures precede and are crucial in perceiving someone else's discomfort or mental state in general. Empirical data reveal that by middle childhood, children are capable of displaying more sophisticated forms of empathy, which are apparent on their interpersonal prosocial behaviors (Abrams et al., 2015). Likewise, research has shown that children's helping behaviors are significantly influenced by the perceived need and merit of the recipient, an ability that also develops with age. For example children allocated more resources to groups of people that have been historically disadvantaged within their society (in Glen et al., 2020), while there were also indications for the pervasiveness of increased empathy and positive attitudes. The researchers argue that interventions which combine contact and social-cognitive programs that promote intergroup empathy are more effective in promoting positive intergroup attitudes.

The Integrated Threat Theory

Intergroup threats raise emotional concerns for the ingroup's welfare (e.g., fears of losing domination), that is, groups which possess the power to inflict or harm the ingroup, constitute a threat to the very existence of the ingroup—*Realistic Threats* (Stephan et al.,

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2009), whereas those holding different values constitute "a threat to the unified meaning system of the ingroup" (p. 44)—*Symbolic Threats*. In this sense, threats posed by the outgroup hold a mediating role between contact and prejudice, since they have the power to inhibit intergroup relations and increase prejudicial attributions. Indeed, attitudes toward refugees are globally deteriorating and negative feelings are also spiraled by the way the outgroup is being projected and communicated by the media (Glen et al., 2020). The majority of European adults, report concerns, perceived threats to the national security and experience anxiety when accepting refugees into their country (De Coninck et al., 2019) and children may be witnessing such concerns within the family context.

Nevertheless, speculating that the content of threats is the same irrespective of age differences and using threats measures interchangeably in studies with children and adult populations may be misleading; the pattern between contact and prejudice through threats may be different for children, adolescents, and adults.

Symbolic Play and Puppet-Mediated Encounters in Intergroup Contact

In their 2019 literature review Kroger and Nupponen present 15 papers that regard school age students and the implementation of puppets as a pedagogical tool, along with empirical evidence indicating associated benefits like generating communication, supporting a positive classroom climate, fostering cooperation and integration into a group and changing attitudes. Kroger and Nupponen (2019) argue that puppets can be beneficial for all children, through specific adaptations and child-sized puppets are more appropriate to promote socially appropriate behaviors through role-playing, since children can more readily attach to them and be intrigued by them. Dunst (2012) also reports benefits in promoting acceptance of children with disabilities and comments: "What is perhaps most encouraging is the fact that such a simple intervention had discernible positive effect, and that the intervention could be delivered to a larger number of students on a single occasion" (p. 455). Ahlcrona (2012) discusses how, in her study, children expressed their thoughts regarding the puppets; "how (they) imagined that the puppet thought, felt, knew, understood, learned, and had ideas and skills" (p. 181), and all this processing is an indicator of children's interest in the mental processes and abstract thinking of other people, that is, empathy and perspective taking. These are exactly the processes that Bruneau et al. (2015) found earlier that directly influence empathy and dilute a strict categorization between "us" versus "them." Empathy induction is a key element for positive intergroup contact, since it promotes inclusion of other in the self, as it underlines the commonalities people share due to their human-like nature and by familiarizing procedures, reduces the threats posed by an outgroup. Puppets in the studies discussed in this section acted as a medium of contact between the ingroup and the targeted outgroup and enhanced the positive feelings toward the targeted outgroup. This idea has been recently used by Peled et al. (2020) who proposed the use of telepresence robots to promote intergroup contact between conflicting groups and achieve prejudice reduction, as a midpoint between online communication and face-to-face meetings. Hence, finding a proper medium in difficult contact situations may be implemented in a variety of manners, and be effectively beneficial, considering children's appeal for symbolic play.

The Present Study

The study implemented a quasi experiment (an intervention) based on the pretest, interaction, posttest (immediate and delayed) design with a control group (Shadish et al., 2002). The intervention took place within a public primary school in Athens and aimed at creating controlled intergroup contact conditions, through empathy inducing activities of cooperation with a child-sized puppet possessing the identity of a newcomer classmate who comes from Syria, and is a refugee (Intervention 1). Children in the Intervention 2 group were merely exposed to the outgroup puppet who was physically present in the classroom, but had no structured, predesigned "interaction" with the classroom; only had a passive presence. The experiment involved two experimental groups and a control group.

Hypotheses of the Study

Hypothesis 1: Children in both intervention groups would hold more favorable attitudes, perceive less threats posed by the outgroup, display increased empathy toward the Syrian refugees, and exhibit more intentions for contact with the outgroup, compared to the control group.

Hypothesis 2: Results in the intervention group that incorporated contact and empathy inducing activities would be more profound compared to the intervention group that was merely exposed to the puppet and would last more in time.

Hypothesis 3: Children in the two intervention groups would show more favorable behaviors toward the Syrian refugee outgroup on the money allocation task, compared to the control group.

Hypothesis 4: Empathy, IOS, and threats would mediate the relationship between contact with the puppet and attitude outcome measures.

Methodology

Participants

The participants in the study were third-grade students (with a M_{age} of 8.9 years in April) and fifth-grade students (with a M_{age} of 10.9 years in April) drawn from a relatively ethnically homogeneous school (65%–70% Greek students) situated in one of the suburbs of Attica. The specific school was chosen based on the fact that: (a) the majority of the students were of a Greek background and (b) that up until the time the study took place, no Syrian refugee children had attended the school. There were three participating third-grade classrooms and three participating fifth-grade classrooms (please see Figure 1). Assignment of the classrooms to each condition was random. Even though the groups were not preequalized in any systematic way (other than age, area they lived, and type of school), between groups comparisons revealed that there were no differences regarding gender proportions between the groups. However, children were not matched in a number of other factors like socioeconomic status (SES), average of achievement, parents' political

Figure 1
The Four Puppets Used in the Intervention Study

Note. From left to right: Hassan, Ali, Bahram, and Asil. See the online article for the color version of this figure.

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orientation, etc. Having said this, the common practice is that students are randomly allocated in the beginning of the year and not on the basis of social background or previous school performance. The total final sample (after parents' consensus was granted) consisted of 83 students (44 boys and 39 girls). However, this number dropped to 73 (please see Table 1) as during the late reassessment phase, five of the children had changed schools, three of the children could not be identified to their pre and postmeasures and two children were absent on the day that the late-measurement phase took place.

Procedures—Description of the Intervention

After being granted approval from the Greek National Bioethics Committee a 4-week intervention was implemented; contact with a child-sized puppet that was given the identity of a Syrian refugee boy of the same age as "his" classmates (the puppets are shown in Figure 1). The teachers read aloud information about the new "classmate's" identity, for example, where did he come from, why he is in Greece and children were told that he was their new classmate, since his "family" had recently moved to Greece due to the war in Syria. The puppet was given a specific seat in the classroom. Prior to the intervention all children had been given a simple comprehensive definition of the refugee status and were administered the premeasures-Questionnaire by their teachers. The teachers read aloud each item to assure a better comprehension of the items, as younger children may have reading difficulties, and were allowed to answer only to enquiries that would refer to unknown words.

Table 1
Number of Participants as a Function of Gender

Type of intervention	Gender		Total
	Girls	Boys	
Intervention 1	9	14	23
Intervention 2	10	14	24
Control group	15	11	26
Total	34	39	73

Experimental Condition 1—Contact and Activities

The puppet in the Intervention 1 group "participated" in a number of predesigned activities, within the context of a curriculum course called Flexible Zone "Evelikti Zoni" (a course that provides content flexibility and is usually implemented through a number of group activities within the classroom). The puppet's participation was achieved with the help of a different group of children each time, that is, a group of four children held the puppet and imitated its voice or carried it along each activity. In general, Intervention 1 aimed at creating cooperation conditions in order to reinforce contact with the puppet through empathy inducing predesigned activities. For example, in the introductory activity, the children were introduced to the "new classmate" through the *Cory and Repeat Game* in which a child gives the signal for saying or doing something and then everybody follows her/him, for example, "Let's all become airplanes," "Let's all laugh really loudly," etc. The game goes on, until everybody had the chance to give their signal. The puppet was held by a different child for each signal. Other activities aimed at cooperation and trust building; Children were given a secret number that only they and the teacher knew. The teacher started telling a story about a group of children who were on a field trip. The children were asked to pretend doing whatever the teacher said, like for example "cross over a small river, climb up a tree" etc. "Then suddenly, number 3 faints . . . ." Number 3 must pretend she/he faints and falls. The task was for the rest of the team to grasp Number 3 before she/he reaches the floor. The children who were "saved" by their classmates took a special seat in the classroom and encouraged their classmates to keep on "saving" children. The game ends when only two children are left. Again, the puppet was carried by different children, every time a new game begun. Activities that aimed at empathy induction were based on role playing, e.g., like the *Extra-terrestrial* activity where a child, either holding the puppet or not, would be the *Extra*-terrestrial who other children try to convince on a matter, for example, to change the kind of food she/he likes, or the friends he/she prefers. The child being the *Extra-terrestrial* is put on a position where he/she needs to support his freedom to express him/herself, to wear the clothes or to choose the friends he/she likes, etc. The extra-terrestrial refuses or does not understand; however, the extra-terrestrial is willing to become friends with the classroom which has to deal with two challenges: To find a way to communicate and also to convince the extra-terrestrial to listen to them. All activities were followed by a group discussion, where the teacher would lead the classroom in reaching outcome(s) in relation to feelings experienced in each instance/role and in relation to the objectives of each activity. The puppet had no other participation during all other in-classroom courses and activities.

Experimental Condition 2—Mere Exposure to the Puppet

In the second experimental condition of the intervention the puppet was only present, sitting on "his" chair "watching" the class. In general, children participating in Intervention 2 only had mere exposure by being in the presence of the puppet but not

interacting with it, at least in a structured predesigned manner—children might only observe the newcomer.

In both intervention groups the teacher moved the puppet from one group of children to another, so as to change seats and sit with a different group of children at the beginning of each week. The puppet was not allowed to be taken outside the classroom during recess time, nor to be removed by children.

Thereby, the defining difference between the two intervention groups regarded the active versus passive presence of the puppet and the perception created to the classmates by this presence; the basic argument being that empathy induction is much easier if there is an active presence, because then the puppet becomes more humanlike. It is then much more likely to include the other in the self, if this "other" is perceived to have similar or quite similar characteristics to oneself.

Control Group

Children in the control group had no direct contact with the puppets and received no specific directions, nor did their teachers. Thereby, they continued with their normal schooling program.

The teachers of the four participating "intervention" classrooms (see Figure 2) were accordingly guided by the researcher, in order to comprise similar methodology and similar activities during the intervention course, that is, allow the participation of the puppet through the preascribed activities in the Intervention 1 groups and allow for mere exposure to the puppet while following their normal schooling program in the Intervention 2 groups.

Measures

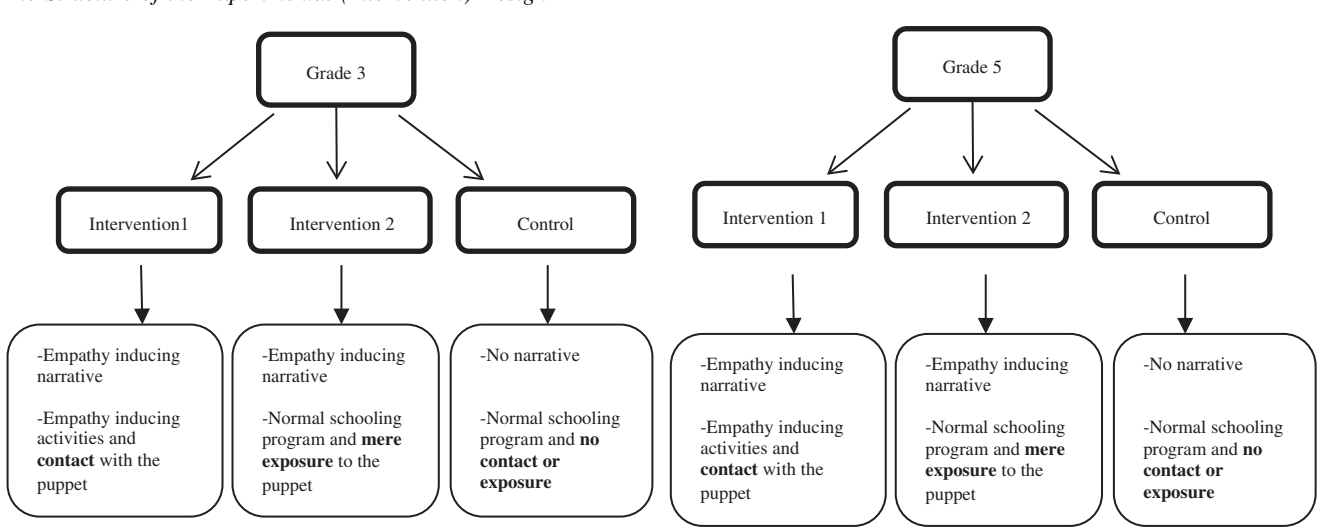
Measures regarding the intervention part of the study were administered in three subsequent occasions, that is, before the intervention took place, 1 week after the intervention had finished (i.e., in 5-weeks-time) and 6 months later as a late response measure (see Figure 1). The timeline of the study is illustrated in Table 2.

Table 2
Timeline of the Intervention

Time period	Type of engagement
Late April 2018	Prequestionnaires (Time 1)
May 2018	4-week intervention (4 × 45')
June 2018	Postquestionnaire (Time 2)
December 2018	Late Response Questionnaires (Time 3)

The measures in the questionnaire included the scales that emerged from a preceding pilot study in order to reveal intergroup threats perceived by children and regarded *Children's Perceived Realistic Threats (CPRT)* including statements like: "There are a lot of jobs in Greece for everyone," "Refugees will throw trash in the city," "There will not be enough space for everyone to live in Greece," measured on a 5-point Likert Type scale ranging from totally disagree to totally agree and *Group-Esteem Threat* with two statements: "Refugees love Greece," "Refugees will like our schools." In addition, two empathy scales were used; a trait empathy scale, that is, *Empathetic Sadness scale* (an adapted version from de Wied et al., 2007) including items like: "Seeing a child who is crying makes me feel like crying," "It makes me sad to see a child who can't find anyone to play with" and Empathy Toward the Refugees (adapted from Turner & Brown, 2008), including two items: "It must be scary for refugees when they arrive in a new country" and "We should be nice to refugees and help them settle in," *IOS* (a graphical, nonverbal measure of the closeness of the self and the ingroup to the outgroup), two *Feeling Thermometers* for children who were born or have been leaving in Greece for a long time and Syrian refugee children, respectively, an *Attitudes toward the Refugees scale* (adapted from Turner & Brown, 2008) including items like: "I like refugees," "Refugees are good people," a *Contact Behavioral Intentions scale*; CBI (adapted from Cameron et al., 2006) which concerned the kind of relationship children would like to have with a future newcomer Syrian refugee child including items like: "I would like to play with him/her," "I would like to sit next to

Figure 2
The Structure of the Experimental (Intervention) Design



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him/her in the classroom" *(please see Cronbach's α for all scales* in Table 3). Finally, a money allocation task adapted from Batson (2010) was implemented, in which children were told that the parents' association had collected the amount of 2000 euros in order to help clubs/organizations in need. The task informed the children that parents' association asked the children to vote as to how they wanted the money to be allocated. Children were presented with five potential competing groups in need, namely an environmental organization, a club for the elderly, a Syrian refugee camp, a dance club, and an organization for the handicaped and were told that they should choose which clubs/organizations to help and then split the money as they wished.

Results

Between Group Comparisons at T1 (Prior to the Intervention)

To investigate any group differences among the three groups prior to the intervention a one-way between-subjects analysis of variance (ANOVA) analysis indicated a successful randomization of participants in the three groups of the intervention, since mean scores on all variables did not differ significantly at the premeasures phase (*Time 1*). Another important finding of the premeasures was a general positivity in all groups toward the Syrian refugee children illustrated by mean scores above the scale midpoint for almost all positive measures and below the scale midpoint for measures of negativity.

Main Analyses

3 (Time) × 3 (Intervention) Mixed MANOVA

A *3 (Time) × 3 (Intervention) Mixed* Multivariate Analysis of Variance (*MANOVA*) design was implemented to investigate within groups differences among the three different times of measurement (pre, post, and late measures). Between-groups differences among the three different groups (Intervention1, Intervention 2, and control group) were also investigated.

Main Effects

Results from the multivariate tests of significance revealed a within subjects significant effect of *Time of Measurement* $\Theta = 0.724$, $F(12, 59) = 3.561$, $p < .001$, $\eta^2 = .438$, of *Type of Intervention* $\Theta = 0.220$, $F(6, 65) = 2.381$, $p < .05$, $\eta^2 = .180$ and of the interaction of *Time* and *Type of Intervention* $\Theta = 0.724$, $F(12, 59) = 3.561$, $p = .001$, $\eta^2 = .420$. All effect sizes were medium to large, since $.11 > \eta^2 > .06$ indicates a medium effect and $.20 > \eta^2 > .14$ indicates a large effect (Cohen, 1988).

Univariate Tests and Interaction Effects

The significant multivariate tests were followed by separate univariate ANOVAs on the outcome variables which revealed significant within-subjects treatment effects of *Time* on *IOS* $F(2, 138) = 7.902$, $p = .001$, $\eta^2 = .103$, on *Group-Esteem Threat* $F(2, 138) = 5.455$, $p = .005$, $\eta^2 = .073$, on *CBI* $F(2, 138) = 3.323$, $p = .039$, $\eta^2 = .046$ and on *Attitudes Toward the Refugee* $F(2, 138) = 5.992$, $p = .003$, $\eta^2 = .080$. In addition, the interaction effect of *Time × Intervention* had a significant effect on *IOS* $F(4, 138) = 4.246$, $p = .003$, $\eta^2 = .110$ and *Empathy Toward the Refugee* $F(4, 138) = 4.338$, $p = .002$, $\eta^2 = .112$. Results are shown in Table 4.

Further pairwise analysis of the mean differences based on *Time* revealed the time points that these differences occurred within each intervention group and are graphically depicted in Figure 3. As illustrated mean differences based on different time points were found to be significant within the Intervention 1 group for IOS when comparing baseline measures (*Time 1*) to post (*Time 2*), showing that post and late measures were significantly more positive compared to premeasures, while there was a significant decrease from *T2* to *T3*. Empathy toward the Refugee was significantly increased in the post and late measures, compared to baseline measures. Finally, within the Intervention 1 group, a significant difference was identified when comparing pre (*T*1) to postmeasures (*T2*), indicating improved attitudes in *Time 2*.

Coming to the Intervention 2 group significant mean differences were found showing increased IOS *from Time 1* to *Time 2*. In addition, increased *Group-Esteem Threat* was found from *Time 2* to *Time 3* and concerning *Attitudes toward the Refugee* a significant improvement was revealed from *Time 1* to *Time 2* and then a significant deteriorating from *Time 2* to *Time 3*.

Regarding the control group significant differences were identified for both *Empathy toward the Refugee* and *CBI* from *Time 1* to *Time 3*. These differences are illustrative of the deteriorating that occurred throughout the time period from the onset of the study to the follow-up measures taken 6 months after the intervention part (for Groups 1 and 2) had finished.

A final remark to be made has to do with that, children in the intervention groups scored higher on all measures that indicate positivity and lower in the measures that indicate negativity toward the refugee at T2 and T3, compared to the control group.

Table 3
Cronbach's α for the Scales in the Study

Time of measurement	CPRT	Empathetic sadness	*r* Empathy refugees	Attitudes	CBI
Pre (Time 1)	.61	.65	.27*	.87	.93
Post (Time 2)	.68	.79	.26*	.91	.94
Late (Time 3)	.62	.67	.34**	.90	.94

Note. Please note that *r* Empathy refugees is not a Cronbach's α value, it is the correlation coefficient between the two items of this construct. CPRT = children's perceived realistic threats. CBI = Contact Behavioral Intentions scale.
* $p < .05$. ** $p < .01$.

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Table 4
Significant F-Tests for Univariate (Within-Subjects) Follow-Up Tests

Effect	Measure	F	η^2	$df1$	$df2$
Time	IOS	7.902**	.103	2	138
	Group-esteem threat	5.455**	.073	2	138
	CBI	3.323*	.046	2	138
	Attitudes toward refugee	5.992**	.080	2	138
Time × Intervention	IOS	4.246**	.110	4	138
	Empathy refugee	4.338**	.112	4	138

Note. IOS = Inclusion of Other in the Self. CBI = Contact Behavioral Intentions scale.
* $p < .05$. ** $p < .01$.

Mediation Analyses

Parallel Mediation analysis was conducted using *PROCESS* (Hayes, 2012) based on Model 4 with bias-corrected percentile bootstrap confidence intervals. The mediators tested were *Empathy Toward the Refugee, Empathetic Sadness, Group-Esteem Threat, and Inclusion of Other in the Self.* Each potential mediator was tested for its impact on two outcome measures separately, namely on *Contact Behavioral Intentions* and on *Attitudes toward the Refugees.* Results after controlling for the impact of the scores at *T1* on the four mediators and of the outcome measure revealed that the only significant indirect effect was that of IOS for Attitudes Toward the Refugees, $b = 0.36$, BCa CI [0.10–0.65], and for Contact Behavioral Intentions, $b = 0.35$, BCa CI [0.11–0.61], while the rest indirect effects were not significant (please see Figures 4 and 5) and this finding was also evident for the Intervention 2 group when compared to the control group; $b = 0.43$, BCa CI [0.16–0.82] on Attitudes Toward the Refugees and $b = 0.51$, BCa CI [0.20–0.88] on CBI.

Assessing the Behavioral Measure

As presented earlier the study incorporated a behavioral measure in *T3* administered 6 months after the intervention had finished. One-way ANOVA did not show any significant differences regarding the amount of money allocated to the refugees compared to the amount of money allocated to other groups in need $F(2, 70) = 2.12$, $p = .128$, nor did for the decisions not to allocate any money for the refugees, compared to allocating money to other groups $F(2, 70) = 1.97$, $p = .147$. Despite this finding, the mean amount of money allocated to the refugees was the highest in the Intervention 1 group (X = 515.22 euros, SD = 341.60), followed by the Intervention 2 group (X = 458.33 euros, SD = 350.36) and then the control group (X = 319.23 euros, SD = 343.825). Even though the difference in the mean amount of money allocated to the refugees in the Intervention 1 group profoundly exceeds the respective amount of money allocated by the control group (i.e., 161.86 euros), which may constitute a large effect, yet this behavioral distinction marginally failed to reach significance when the Intervention 1 group was compared to the control group, $F(1, 70) = 3.93$, $p = .051$.

Though the one-way ANOVA test came out marginally nonsignificant, planned contrasts were performed, based on that the study included a control group. Accordingly, a one-tailed significance value was applied, based on the prestated hypothesis that the intervention groups will exhibit more positive behavior. Planned contrasts revealed a significant difference between *Intervention 1* group and the *Control* group with $t(70) = 1.983$, $p = .013$ (one-tailed), $r = .22$ (small effect).

Discussion

The study investigated the prevalence of children's attitudes in Greece toward the Syrian refugee children, whom they had never encountered with by the time the study was taking place and up until the intervention had finished. Findings of this study are important because they show that the application of ideas from the pioneering work of Bruneau et al. (2015)—where an emphasis in the mental characteristics of out groupers can promote empathy and dilute strict "us" versus "them" categorizations—are verified in the classroom context in a real-life situation in Greece.

The aim of the study was to investigate the effectiveness of a new form of intergroup contact; contact with a puppet, as well as to verify the mechanisms through which intergroup attitudes may improve, along with intentions for intergroup contact and even intergroup behaviors among children. The study showed how contact with a puppet, under optimum conditions within the classroom context resulted in more positive attitudes toward the outgroup represented by the puppet. The study also showed how indirect techniques to promote empathy and IOS are effective in reducing prejudice for targeted outgroups and this was done through a design that involves measures at a distance of some months, which is rarely incorporated (Paluck et al., 2019). In addition, the study examined the potential mediating role of perceived intergroup threats, which is another challenge in studies with children, since measures of threats are meant to address adult populations.

In specific, results partly confirmed Hypotheses 1 and 2, since some of the assumed constructs were not improved for neither of the two intervention groups, whereas any observed improvements were more profound in the intervention group that involved contact and activities.

Regarding Hypothesis 1 specifically, none of the two intervention groups showed significantly increased empathetic sadness compared to the control group. In addition, the intervention group with mere exposure to the puppet did not show increased levels of *Empathy toward the Refugees*, compared to the control group. Thus, only the intervention group with contact and empathy inducing activities had significantly increased empathy toward the refugees at *T2,* right after the intervention had finished. Most importantly, children in the Intervention 1 group continued to exhibit equally high levels of empathy toward the refugees, 6 months after the intervention had finished, as opposed to the other two groups whose levels of empathy toward the refugees had dropped to the original levels prior to the intervention for the Intervention 2 group and even below the premeasure for the control group. Nonetheless, children in the two intervention groups scored significantly lower in *Group-Esteem Threat.* In addition, both intervention groups scored significantly higher on the *IOS,* the *CBI,* and the *Attitudes toward the Refugee measures*, while children that had intergroup contact with the puppet, along with activities, scored significantly higher on the *IOS* measure, compared to children that only had mere exposure to the puppet.

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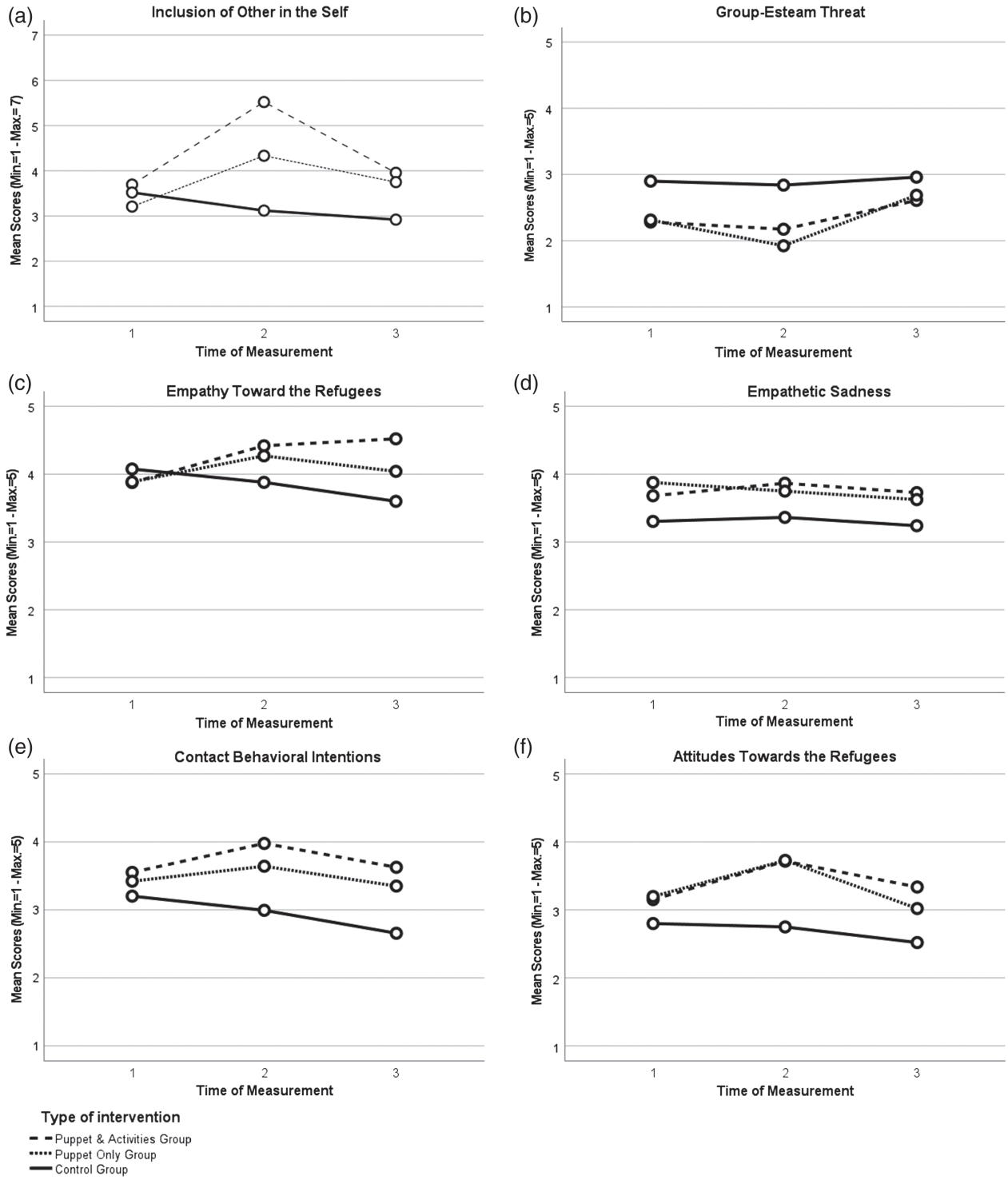
Figure 3

Group Differences at T1 (Pre), T2 (Post), and T3 (Late) Measures



Note. Figure 3a has a scale from 1 to 7, whereas all other scales range from 1 to 5.

Altogether results from measures on *T2* give support to Hypothesis 2 which proposed that intergroup contact with activities will be more effective compared to mere exposure to the puppet. This finding is in line with Pettigrew and Tropp (2008) who argue that structured intergroup contact results in significant greater effects. Hypothesis 2 is also indirectly supported by the fact that, even though mean differences between the two intervention groups were not significant, still children in the Intervention 1 group scored

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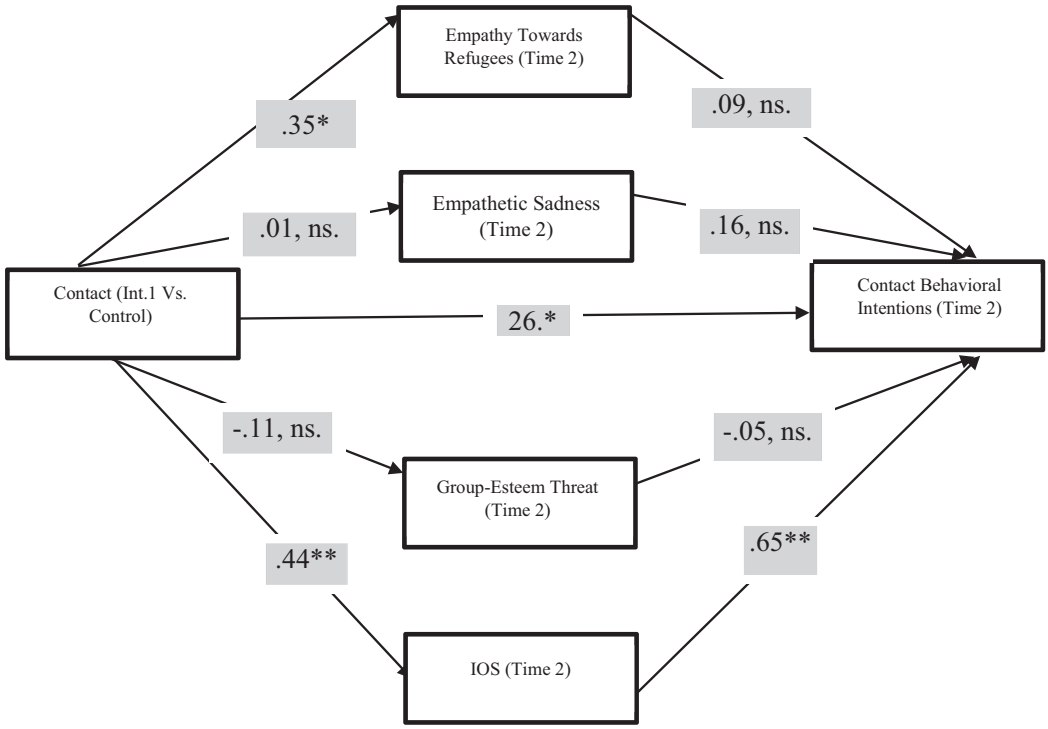
Figure 4

Longitudinal Parallel Mediation with 4 Mediators in the Comparison Between Intervention 1 Group and the Control Group on Contact Behavioral Intentions



Note. The longitudinal mediation model embodied within the quasi experiment—Comparison of the impact of Intervention 1 (Contact and Activities) with the Control Group on Attitudes Toward the Refugees through Empathy Toward the Refugees, Empathetic Sadness, Group-Esteem Threat, and Inclusion of Other in the Self (IOS).

higher, compared to the Intervention 2 group, on all measures that indicate positivity toward the refugee and lower in the measure that indicate negativity toward the refugee (i.e., Group-Esteem threat). The critical element for these findings is probably related to the "active" role of the puppet in the Intervention 1 group. Children in this group would carry the puppet across the classroom in order for him to "participate" in the various activities, while at times, children carrying the puppet became the "cyranoid" between the puppet and the rest of the classroom, by speaking the puppets' voice, wills and even emotions, an ability which requires empathetic skills. As Remer and Tzuriel (2015) argue what makes the puppets unique as a teaching tool is to be found in their ability to "move" and "speak," as if real humans, and these abilities elicit identification by people "who can see themselves or part of themselves in the puppets" (p. 356). On the other hand, children also benefited from mere exposure to the puppet, even though at a lesser degree. This finding is in line with Crisp et al. (2009) findings on extended contact who argue that the idea of contact alone becomes "mentally articulated in the form of knowledge about experiences of others, that has an impact on intergroup attitudes" (p. 1) and that this knowledge does not need to be confirmed through actual experience to result to projection to outgroups (Stathi & Crisp, 2008). Likewise, the mere presence of the puppet in the classroom could have created a sense of inclusivity as classmates, even if mere presence would not offer possibilities for empathy, offered by the other experimental condition.

Overall, short-term benefits were identified within both intervention groups concerning *Inclusion of Other in the Self (IOS)* and on *Attitudes toward the Refugees.* Coming to the very important long-term effects, an increase in *Empathy toward the Refugee* was identified within the Intervention 1 group 6 months after the intervention had finished. Likewise, there were indications of the buffering effect within both intervention groups for almost all measures.

As it comes to Hypothesis 3, children in the two intervention groups showed more favorable behaviors toward the Syrian refugee outgroup on the money allocation task, compared to the control group, and this positivity was more profound for the intervention group that engaged through activities with the puppet, even though this was marginally statistically nonsignificant. However, small samples may impact the statistical significance between two groups, resulting in a nonsignificant p value, even when this difference produces a large effect size (Field, 2013). Vezzali et al. (2014, 2017) as well as Glen et al. (2020), also found trends of more positive behavior coming from the intervention groups in their study with children (2 weeks after their intervention had finished), which did not, however, reach statistical significance. Finally, in relation to Hypothesis 4, only *IOS* mediated the relationship between contact and attitudes/intentions. In fact, *IOS* appears to be a strong and consistent mediator, since its mediating role was revealed for both intervention groups when compared to the control group and after controlling for measures prior to the intervention.

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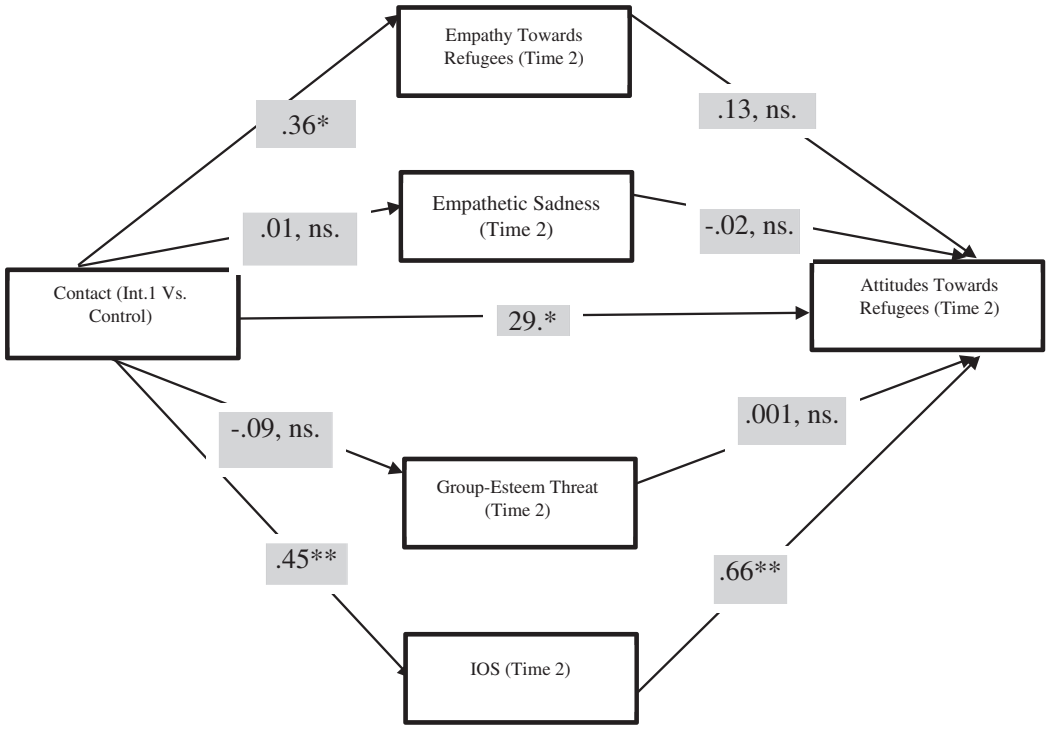
Figure 5

Longitudinal Parallel Mediation with 4 Mediators in the Comparison Between Intervention 1 Group and the Control Group on Attitudes Towards Refugees



Note. The longitudinal mediation model embodied within the quasi experiment—Comparison of the impact of Intervention 1 (Contact and Activities) with the Control Group on Contact Behavioral Intentions through Empathy Toward the Refugees, Empathetic Sadness, Group-Esteem Threat, and Inclusion of Other in the Self (IOS).

Altogether, results showed that contact with (and exposure to) a puppet that possesses the identity of a refugee Syrian boy improved primary school children's attitudes, and intentions and also produced more favorable actual behaviors toward Syrian refugee children. Even though intergroup contact with empathy inducing activities produced the most profound positive outcomes, the most striking finding is that improvements were also detected, though at a lower degree, even when children were only exposed to the puppet. This finding implies that mere exposure to the puppet was enough to result in familiarizing with the outgroup, and this was also evident in the reduction of the Group-Esteem threat posed by the refugee Syrian children from T1 to T2 measures. This last finding is compatible with the well-established positive relation between intergroup anxiety and prejudice, in that a decrease in anxiety (via reduced intergroup threats) leads to prejudice reduction. This finding shows that with minimum means and effort (an easy to apply intervention), significant results can still occur.

The positive trends identified within both intervention groups at *T2* and most importantly the buffering effect that held the deterioration of attitudes in both intervention groups at *T3,* that is, at a time of a general deterioration in Greece and globally (Glen et al., 2020), which was, on the other hand, illustrated in the deepest deterioration within the control group, is a significant and promising finding. In essence, within the control group results on *Empathy toward the Refugees* and on *Contact Behavioral Intentions,* became significantly worse from *T1* to *T3,* and this finding is likely to reflect the general negativity that was growing within the Greek society, as the refugee issue had become a major political arena between the government and right political parties and weekly protests in rural Greece against refugee relocation were frequently taking place (Kirtsoglou & Tsimouris, 2018). Under the prism of such negative changes promoted within the societal channels in Greece, achieving long-lasting benefits for empathy toward the target outgroup and restraining attitudes deterioration for both intervention groups, highlights the effectiveness of the intervention in promoting coexistence and peace.

Contribution of the Present Study

The intervention in the present study incorporated a novelty in the field of intergroup contact interventions by implementing direct contact with a puppet. By doing so, the intervention combined features of face-to-face contact, as well as features of imagined contact, for children had to mobilize their imagination to interact with the puppet. In doing so, the study overcame a traditional barrier in implementing contact interventions within the educational context, that is, the difficulty to bring together members of diverse groups, either because they live physically segregated or because they are members of opposing groups (Hewstone et al., 2014; Liebkind et al., 2018), while it guaranteed for optimum conditions

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as described by Allport. The intervention created "intergroup interactions" with an unfamiliar outgroup that would soon become part of the children's everyday reality, thus prepared the children for future contact and offered valuable first-hand knowledge to them, only without running the risk of intergroup anxiety.

In addition, the present study proves that interventions in real-life context can be both achievable and beneficial, and this is very important regarding social psychology in general, since many studies have been carried out in rather benign settings (e.g., contact on college campuses) and a "major thrust of research in social psychology has been to supply evidence from more demanding settings" (Hewstone et al., 2014, p.41).

The present study is also important in that it provides evidence of the ease to successfully induce empathy among children, at a time of strong indications of deterioration of humanitarian values and principles (Krznaric, 2015) and highlights the necessity to incorporate such interventions in schools.

Another important feature of the intervention is that it was classroom-led by teachers, and not researchers, within the context of the official curriculum, thus it was easily and readily implemented, and can be repeated as such by teachers and/or other practitioners.

Limitations and Future Directions

Despite the originality of this study, there were also a number of limitations which need to be addressed. Starting from the promising fact that it was delivered by teachers, we must acknowledge the potential confounds this brings into play, since teachers' implicit bias, expectations, motivation, level of engagement, even competence in carrying out an intervention vary. Thus, even though interventions delivered by teachers offer ecological validity, yet they are always more exposed to teachers' subjectivity, while they usually produce weaker effects (Ülger et al., 2018). Future interventions need to train teachers in order to counterbalance for the lack of strict experimental control, since, despite the problems yield, theoretically driven field experiments carried out by teachers stand more chances of being incorporated in the school context, since they can be performed on a collective and costless manner.

Coming to more methodological issues the quasi experimental design of the study naturally limits arguments for causal inferences and this is further weakened by the absence of another experimental condition that would implement interaction with the puppet by using "neutral" activities, that is, not aiming to induce empathy, in order to provide a clearer depiction of the impact of empathy and IOS on attitudes/behaviors. In addition to this point, the between groups comparisons at Time 2 showed the impact of the interventions in the study as measured right after the intervention had finished. However, since children of the control groups were in the same school with the intervention groups, there is a possibility that children from the control groups may have inadvertently come into contact with the puppets, extended, or direct. This possibility would result in masking the outcomes of the study, since children in the control groups may have also improved their attitudes toward the specific outgroup. Future studies could overcome this limitation by having two control groups; one in the school that the intervention is taking place and a control group in another school with a matched sample. The second point is that the empathy measures used in the present study may not have adequately addressed the level or type of empathy experienced by participants. As Glen et al. (2020) discuss: "capturing the complexity and multidimensional experience of empathy may provide a clearer picture of its role in attitude formation and outgroup-directed prosocial behavior in the future" (p. 82). Indeed the activities used in the Intervention 1 group aimed at promoting empathy through a sense of identification with the puppet that one "handles" and which happens to be a refugee, rather than being directed toward the specific outgroup. In this sense the intervention of the study is "minimal" and could be furthered enhanced in future research. In short, had the activities been specifically tailored to the refugee experience, for example, by role playing a story where the coastguard sees a young refugee in open sea on a small boat, the impact of the intervention might have been stronger.

Another limitation of the study regards the relatively low reliability of the Empathetic Sadness scale, as well as the moderate correlation between the two items used on the Empathy toward the Refugees scale. Concerning the Empathetic Sadness scale, Cronbach's α reported by de Wied et al. (2007) varied from .68 to .76 (seven items scale). Given that the scale was used as a 6-item scale in this study, along with the fact that the number of items impacts Cronbach's α, the corresponding values reported in this study are acceptable. Concerning the 2-item Empathy toward the Refugees scale, the reported r coefficient indicates a moderate correlation, which is similar to Turner and Brown (2008) respective values (r = .34, r = .41). Nonetheless, future studies could implement bigger sample sizes or use more comprehensive measures of Empathy.

The issue of the small sample size combined with the number of tests conducted is also related to higher possibilities of *Type 1 Error* Inflation. A bigger sample size would allow for more statistical analyses to be conducted with certainty (e.g., within group analyses based on age or other grouping criteria) and would address the issue of power analysis. Thereby, the findings of our study are susceptible to the related problems caused by small samples such as inflamed Type I Error and lack of generalizability to the population. Data from a larger number of schools could also be employed so as to cross-validate the current findings and also to allow for additional comparisons, for example, to split the schools and have the interventions being carried out by researchers in half of the schools and by teachers in the other half. Another issue to consider is that the research being mainly quantitative in nature, did not account for qualitative information, neither did for potentially valuable interfering elements like parents' opinion on the matter. Likewise, there was no control over how much access students had in media reports regarding the refugees, and this variable may have confounded the results (Liebkind et al., 2018). Accordingly, there might had been other events occurring, over the 6 months between April and December 2018, that may have influenced the results (e.g., local events, major stories on the news regarding refugees). Future studies could include more covariates that have been known to influence intergroup attitudes, such as students' individual degree of identification with the ingroup, measures of intercultural tolerance climate, in-group norms, parent's political orientation, etc.

Further research that will combine qualitative data and quantitative evidence is, thereby, needed so that to acquire a more holistic depiction of the phenomena under study, and provide first-hand explanations regarding children's reactions to contact with puppets of specific outgroup identities (national or other). An additional potential confound in the findings regards the gender of the puppet,

since all four puppets were boys' figures. The reason for this choice was that studies show that boys are less likely to be victimized by their peers, compared to girls (Spears Brown & Bigler, 2004), and this possibility had to be ruled out, since it would raise serious concerns for the effectiveness of the intervention. Nevertheless, future studies, should focus on finding a proper way by altering the research design and incorporate gender-matched puppets, to rule out the potential intervening of gender stereotypes.

Another future objective is to introduce multiple-component interventions that have been proven to be more effective compared to unidimensional ones (Beelmann & Heinemann, 2014). Vezzali et al. (2017) argue that coupling a direct contact intervention with an imagined contact intervention can be very different from running the two interventions separately.

Altogether the present study found evidence of the impact of mere exposure to a puppet at improving attitudes toward the Syrian refugee and intentions for future contact with the specific outgroup. When contact was implemented via empathy induction activities, positive outcomes were more profound. More importantly, the study found evidence of real behavioral positivity among primary school children, and this is a very promising finding, since it responses to a fundamental request by scholars within the social sciences, that is, finding ways to achieve behavioral changes over and above attitude changes, which is mostly directed to actual peace-building efforts.

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